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## Unit 5 : Session 12 : SLO 2 | LDM/STM, Programmed I/O and Encoding

### I. Multiple Choice

1. B) LDM Load Multiple registers.
2. C) Polling programmed I/O repeatedly checks the device status register.
3. D) DB (Decrement Before) STMFD = Store Multiple, Full Descending stack = decrement-before addressing.

### II. Fill in the Blanks

1. status program-controlled I/O repeatedly checks the device status register.
2. stack LDMFD restores registers from the (full descending) stack.
3. operand binary encoding identifies the opcode and operand fields.

### III. True or False

1. TRUE polling burns CPU cycles continuously checking the device.
2. FALSE LDMIA loads in increasing address order (Increment After).
3. TRUE STR R1, [R2] stores R1's value at the address in R2.

### IV. Match the Following

- A) STMFD 2) Store multiple registers.  
B) LDMFD 3) Load multiple registers.  
C) Polling 1) Waiting for device ready.  
D) STR 4) Write to memory.

### V. Numerical Answers

1. MOV R1, #5 cond=1110, I=1, opcode=1101 (MOV), S=0, Rn=0000, Rd=0001, immediate=0x005 0xE3A01005.
2. 20 bytes 5 registers — 4 bytes.
3. Polling loop:
  - LDR R2, =0x60000020
  - loop: LDR R0, [R2]
  - TST R0, #1 @ test bit 0
  - BEQ loop @ keep waiting until bit 0 = 1